MUHAMMAD NASIF

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PROFESSIONAL SUMMARY

Software engineer specializing in scalable web applications and distributed backend architecture. Experienced in turning ideas into products, from designing to shipping working software. Hands-on experience with Microservices, Micro Frontends, Containerization, Container Orchestration, Deployment pipelines, and Cloud infrastructure management. Passionate about clean architecture, performance optimization, and building production-grade applications.

TECHNICAL SKILLS

- Languages: JavaScript, TypeScript, SQL
- Backend: Node.js, Express.js, REST APIs, GraphQL, gRPC
- Frontend: React.js, Next.js, React Native, Module Federation, Tailwind CSS, PWA
- Architecture: Microservices, Micro Frontends, Clean / Hexagonal Architecture, Modular Monolith, Event-Driven Architecture, Domine-Driven Design (DDD), SOLID, Repository Architecture, MVC Architecture.
- Databases: PostgreSQL, Prisma ORM, MongoDB, Redis, Supabase
- Realtime & Messaging: Kafka, RabbitMQ, Socket.IO, WebRTC, Redis Pub/Sub, BullMQ
- Cloud, DevOps & Observability: Docker, GitHub Actions (CI/CD), AWS (EC2, ECR, RDS, S3, Amplify, CloudFront, OIDC, SSM), Nginx, Grafana, Prometheus, Loki, OpenTelemetry, Doppler
- Testing: TDD, Jest, Mocha, Chai, Sinon, Cyperes
- AI & LLM Engineering: LangGraph, LangChain, Model Context Protocol (MCP), RAG Pipelines, In Process Embeddings, pgvector
- Tools: Git, GitHub, Postman, Figma, Agentic ai tools.

PROJECTS

Voymate – Real-Time Solo Traveler Matchmaking & Community Expedition Platform: Live Link

- Built a full-stack platform for solo travelers to match with trip companions, find nearby travelers in real time, and book verified community trips across a 3-role RBAC system (User, Organizer, Admin).
- Architected a Node.js 22 and TypeScript backend using Clean/Hexagonal Architecture, SOLID principles, and TSyringe dependency injection, paired with a feature-sliced Next.js frontend using TanStack Query and Axios interceptors for automatic JWT refresh and request retry.
- Built location-based traveler matching using PostgreSQL/PostGIS spatial queries and multi-factor preference scoring, alongside a trip booking workflow with Razorpay payment verification and real-time location sharing on Google Maps.
- Integrated a stateful LangChain/LangGraph travel assistant using pgvector for semantic search with local CPU embeddings, and configured BullMQ with Redis for background job queues, OCR document verification via Google Document AI, and response caching.
- Implemented real-time 1:1 chat and P2P WebRTC audio/video calls using Socket.io for signaling and online presence tracking.
- Containerized the backend with Docker and automated CI/CD deployments via GitHub Actions to AWS EC2 (ECR, RDS, S3, Doppler secrets) and AWS Amplify.
- Git Hub Repo: Api, Client

Kafla - Collaborative Group Travel Workspace Live Link

- Architected a 4-layer Clean Architecture modular monolith in Node.js, Express, and TypeScript backed by PostgreSQL 16 (Prisma) and Redis for session and state caching.

- Built a Next.js API proxy (/api/routes) for Google Routes API v2 with response field masking and GPS speed/distance validation using the Haversine formula, alongside httpOnly cookie JWT rotation with CSRF protection.
- Built an AI itinerary planner using LangGraph state machines and a custom Model Context Protocol (MCP) server that enriches recommendations with Google Places data and automatically retries failed schema validations to guarantee valid JSON output.
- Implemented group expense splitting using a greedy minimum-cash-flow algorithm to minimize settlement transfers, paired with Socket.io and WebRTC signaling for live group chat and video calls.
- Built live route mapping and GPS location tracking using the Google Routes API to render multi-waypoint polylines alongside browser Geolocation and Open-Meteo weather forecasts.
- Configured production deployment and CI/CD using multi-stage Docker builds, Nginx reverse proxy (SSL/WebSockets), AWS S3 presigned uploads, and GitHub Actions for automated image publishing and EC2 rollouts with Prisma migrations.
- Git Hub Repo: Api, Client

Gracia- E-Commerce Platform Live Link

- Designed a layered Service-Oriented Architecture in Node.js, Express, and React 18 (Redux Toolkit / RTK Query) using MongoDB ACID transactions for race-condition-free inventory updates, stacked product and coupon discount logic, and streaming PDFKit invoices and ExcelJS reports.
- Integrated Razorpay (HMAC-SHA256 webhook verification), payment retry handling for failed checkouts, guest-to-account cart merging on login, and instant session revocation via Socket.io.
- Built a Gemini-powered shopping assistant that parses user search intent and queries the live product database to recommend matching items.
- Automated Docker deployment pipelines using GitHub Actions CI/CD across AWS EC2, S3, and Amplify.
- Git Hub Repo: Api, Client

Concept Ladder AI Live Link

- Built a full-stack AI learning app that generates topic explanations across three difficulty levels (Kid, Beginner, Advanced) using the Gemini API and React, implementing regex sanitization, JSON validation, and exponential backoff retries to handle malformed LLM responses reliably.
- Integrated Clerk authentication with Supabase PostgreSQL and configured Row-Level Security (RLS) policies to isolate user data.
- Git Hub Repo: Client

EDUCATION

1. **Master of Computer Applications (MCA)**
   IIT Patna | 2026 – Present (Hybrid Mode)
2. **Bachelor of Computer Applications (BCA)**
   IGNOU | 2022 – 2025